

Robustness of State-Space Control Systems

Will the design survive a plant that differs from the model?

Ogata, *Modern Control Engineering*, Ch. 10.

In this tutorial you will:

- form the sensitivity S and complementary sensitivity T of a loop,
- read classical gain/phase margins at the plant input, and
- test the closed loop against plant-parameter perturbations.

Step through with **Ctrl+Enter**, or render a report with **publish**.

A controller designed from a nominal plant model A, B, C must still perform acceptably when the real plant differs (parameter uncertainty, unmodeled dynamics). **Sensitivity** and **gain/phase margins** quantify this robustness for a state-feedback loop broken at the plant input.

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Nominal Plant and LQR Design

```
A = [0 1; -2 -3];
B = [0; 1];
C = [1 0];
D = 0;

Q = diag([5 1]);
R = 1;
K = lqr(A,B,Q,R);
fprintf('Nominal LQR gain K = '); disp(K)
fprintf('Nominal closed-loop poles: '); disp(eig(A-B*K))
```

```
Nominal LQR gain K =      1.0000      0.4641
```

```
Nominal closed-loop poles:   -1.7321   -1.7321
```

Loop Gain and Sensitivity Function

Breaking the loop at the plant input, the loop transfer function is $L(s) = K(sI - A)^{-1}B$. The **sensitivity function** $S(s) = (1 + L(s))^{-1}$ measures how much output disturbances or plant perturbations are attenuated by feedback; the **complementary sensitivity** $T(s) = L(s)(1 + L(s))^{-1} = 1 - S(s)$ measures reference/noise transmission. A small $\|S\|_\infty$ near crossover indicates good disturbance rejection; $S + T = 1$ is an unavoidable design tradeoff.

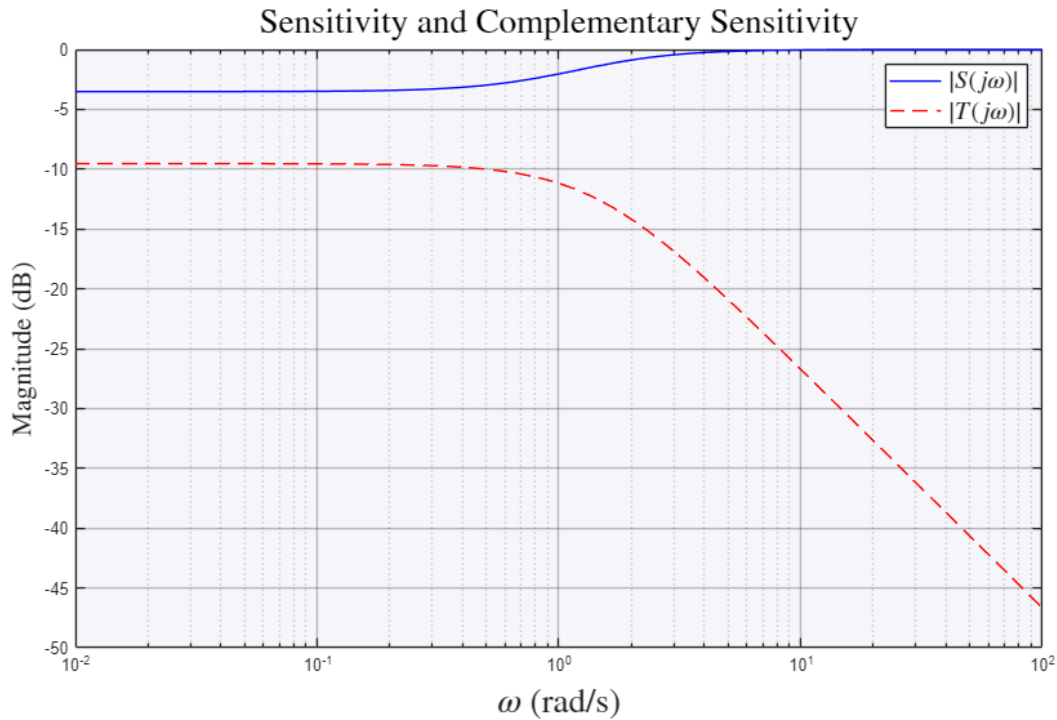
```
sys_ol = ss(A,B,K,0);
L = tf(sys_ol);
S = feedback(1,L);
T = feedback(L,1);

w = logspace(-2,2,500);
[magS,~] = bode(S,w);
[magT,~] = bode(T,w);
```

```

figure
semilogx(w,20*log10(squeeze(magS)), 'b', w, 20*log10(squeeze(magT)), 'r--')
legend('$|S(j\omega)|$', '$|T(j\omega)|$', 'Interpreter', 'latex', 'FontSize', 14)
title('Sensitivity and Complementary Sensitivity', 'Interpreter', 'latex', 'FontSize', 20)
ylabel('Magnitude (dB)', 'Interpreter', 'latex', 'FontSize', 16)
xlabel('$\omega$ (rad/s)', 'Interpreter', 'latex', 'FontSize', 20)
grid on

```



Classical Margins at the Plant Input

```

[Gm,Pm,Wcg,Wcp] = margin(L);
fprintf('\nGain margin = %.2f dB, Phase margin = %.2f deg\n', ...
    20*log10(Gm), Pm)
fprintf('(LQR guarantees PM >= 60 deg, infinite upward gain margin,\n')
fprintf(' and >=50%% downward gain reduction tolerance.)\n')

```

```

Gain margin = Inf dB, Phase margin = Inf deg
(LQR guarantees PM >= 60 deg, infinite upward gain margin,
 and >=50% downward gain reduction tolerance.)

```

Robustness to Parameter Perturbation

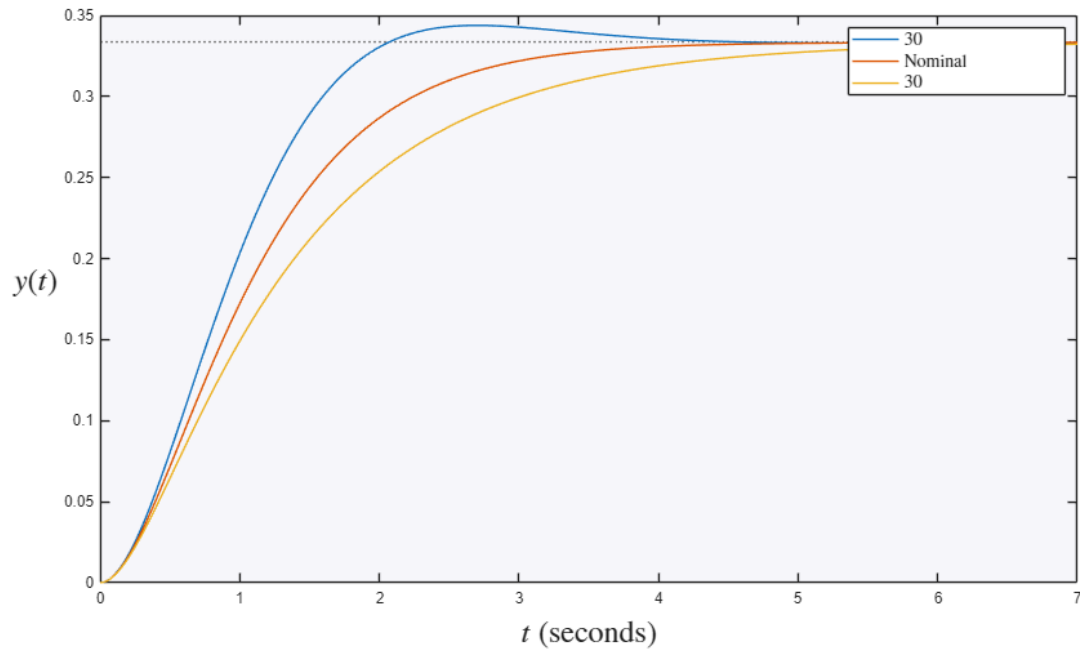
Test closed-loop stability when the true plant's A matrix deviates from the nominal model (e.g. an underestimated damping/stiffness term), while still using the gain K designed from the nominal model.

```

perturbations = [0.7 1.0 1.3]; % multiply the (2,2) damping entry
figure
hold on
for p = perturbations
    A_true = A; A_true(2,2) = A(2,2)*p;
    sys_true_cl = ss(A_true-B*K, B, C, D);
    step(sys_true_cl)
end
hold off
legend('30% less damping', 'Nominal', '30% more damping', 'Interpreter', 'latex', 'FontSize', 12)
title('Closed-Loop Step Response Under Plant Perturbation', 'Interpreter', 'latex', 'FontSize', 20)
ylabel('$y(t)$', 'Interpreter', 'latex', 'FontSize', 20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter', 'latex', 'FontSize', 20)

```

Closed-Loop Step Response Under Plant Perturbation



Stability Margin Check Across Perturbations

```
for p = perturbations
    A_true = A; A_true(2,2) = A(2,2)*p;
    is_stable = all(real(eig(A_true-B*K))<0);
    fprintf('damping x%.1f: closed-loop poles = %s, stable = %d\n', ...
        p, mat2str(eig(A_true-B*K)',4), is_stable)
end

damping x0.7: closed-loop poles = [-1.282-1.165i -1.282+1.165i], stable = 1
damping x1.0: closed-loop poles = [-1.732 -1.732], stable = 1
damping x1.3: closed-loop poles = [-0.8549 -3.509], stable = 1
```

Try it yourself

- Widen the perturbation range (say 0.5 to 1.5) and find where the closed loop finally goes unstable.
- Retune Q and compare the sensitivity peak $\max(\text{magS})$ – a lower peak means more robust disturbance rejection.